

1. $\Delta H > 0$

Endothermic

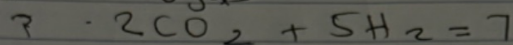
Add Heat

The Temperature decreases
More energy is required to break reactant bonds
than is released forming product bonds

1b $\Delta H = \Delta U + \Delta n_g RT$

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$R = 8.314 \text{ J/mol}\cdot\text{K}$



$\Delta n_g = 7$

$\Delta n_g RT = 7 \times 8.314 \times 298 = 17342 \text{ J} = 17.34 \text{ kJ}$

$\Delta U = 69.36 - 17.34 = 52.02 \text{ kJ/mol}$

1c

$Q = \Delta U$

$m = 64.1 \text{ g/mol}$

$n = \frac{150}{64.1} = 2.34 \text{ mol}$

$Q = n \Delta U = 2.34 \times 52.02 = 121.71 \text{ kJ}$

20/20

1. $\Delta H > 0$

2)

HW

$$q_{\text{reaction}} = -q_{\text{water}}$$

$$q = m C \Delta T$$

$$m = 650 \text{ g}$$

$$C = 4.184 \text{ J/g}^\circ\text{C}$$

$$\Delta T = 36.2 - 23.6 = 12.6^\circ\text{C}$$

$$q_{\text{water}} = 650 \times 4.184 \times 12.6 = 34.28 \text{ J}$$

$$q_{\text{rxn}} = -34.28 \text{ kJ}$$

$$\frac{-34.28}{2.7} = -12.7 \text{ kJ/g}$$

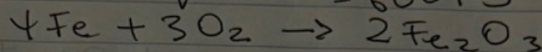
15/20

X
5

3)

$$100 \text{ kJ/h for 6 h}$$

$$q_{\text{total}} = 600 \text{ kJ}$$



$$\Delta H = -1650 \text{ kJ per 4 mol Fe}$$

$$\frac{600}{1650} \times 4 = 1.45 \text{ mol Fe}$$

$$1.45 \times 55.85 = 81.09 \text{ g Fe}$$

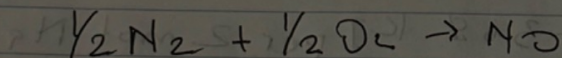
$$\frac{3}{4} \times 1.45 = 1.09 \text{ mol O}_2$$

$$1.09 \times 32 = 34.9 \text{ g}$$

17/20

1. charge
m

49



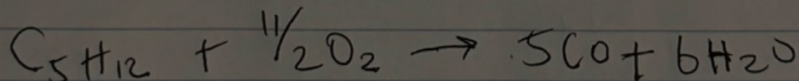
$$\text{NO} = 90.25 \text{ kJ/mol}$$

$$\text{N}_2, \text{O}_2 = 0$$

$$\Delta H = 90.25 \text{ kJ/mol}$$

X-2

4b



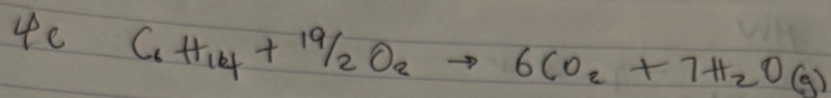
$$5(-110.5) + 6(-285.8) = -552.5 - 1714.8$$

$$= -146.7$$

$$\Delta H = -2267.3 - (-146.7)$$

$$= -2120.6 \text{ kJ/mol}$$

✓



$$6(-393.5) + 7(-241.8) = -2361 - 1692.6$$

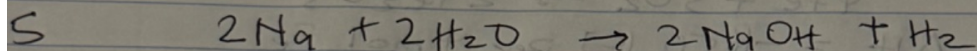
$$= -4053.6$$

$$\Delta H = -4053.6 - (-198.7) = -3854.9 \text{ kJ/mol}$$

4d $-364 + 4(-393.5) = -364 - 1574 = -1938$

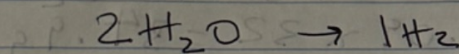
$$-1387 + 4(-110.5) = -1387 - 442 = -1829$$

$$\Delta H = -1938 - (-1829) = -109 \text{ kJ/mol}$$



$$\frac{100}{23} = 4.35 \text{ mol}$$

$$\frac{1}{18} = 0.0556 \text{ mol}$$



$$H_2 = 0.0556 / 2 = 0.0278 \text{ mol}$$

$$\Delta H = -368 \text{ kJ per 2 mol Na}$$

$$0.0556 \text{ mol } H_2O \Rightarrow 10.2 \text{ kJ}$$

Explanation?

15/20